

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
ICSE CLASS 10 - CHEMISTRY

Chapter: Organic Chemistry

Date: _____

Name: _____

Maximum Marks: 20

Time: 45 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. Which pair represents chain isomers?

- (i) Pentane and 2-methylbutane
(k) Ethane and propane

- (j) But-1-yne and but-2-yne
(l) Propene and propyne

Answer: _____

2. An open-chain hydrocarbon with formula C_6H_{10} belongs to the series:

- (i) C_nH_{2n+2}
(k) C_nH_{2n-2}

- (j) C_nH_{2n}
(l) C_nH_{2n+1}

Answer: _____

3. The IUPAC name of $CH_3-CH(CH_3)-CH(Br)-CH_2-CH_3$ is:

- (i) 3-bromo-2-methylpentane
(k) 2-methyl-3-bromopentane

- (j) 3-bromo-4-methylpentane
(l) 3-bromo-2-methylhexane

Answer: _____

4. Which pair can be successive members of the same homologous series?

- (i) C_3H_8 and C_4H_{10}
(k) C_4H_8 and C_5H_{12}

- (j) C_3H_6 and C_4H_{10}
(l) C_2H_2 and C_3H_6

Answer: _____

5. Which functional group is present in alcohols?

- (i) -CHO
(k) -OH

- (j) -COOH
(l) -CO-

Answer: _____

6. $CH_3-CH=CH-CH_3$ is named:

- (i) But-1-ene
(k) Butane

- (j) But-2-ene
(l) But-2-yne

Answer: _____

7. Compounds with the same molecular formula but different structural formulae are called:

- (i) Homologues
(k) Radicals

- (j) Isomers
(l) Catalysts

Answer: _____

8. Which hydrocarbon generally undergoes addition reactions?

- (i) Alkane
(k) Saturated hydrocarbon only

- (j) Alkene
(l) Methane only

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Give the IUPAC name of $\text{CH}_3\text{-C}(\text{Cl})(\text{CH}_3)\text{-CH}_2\text{-CH}(\text{Br})\text{-CH}_3$. [2]

10. Give the IUPAC name of $\text{CH}_3\text{-CH=C}(\text{CH}_3)\text{-CH}(\text{Cl})\text{-CH}_3$. [2]

11. Write the structural formula of 3-bromo-2-methylpentane. [2]

12. Draw or write the condensed structural formulae of all three chain isomers of C_5H_{12} . [2]

13. State the homologous series and general formula for C_4H_{10} , C_4H_8 and C_4H_6 . [2]

14. Why do alkanes generally undergo substitution reactions, while alkenes and alkynes undergo addition reactions? [2]
